DSA 5

Binary Heaps

Complete Binary Tree

Min Heap

Max Heap

NOTE: You are allowed to access only the root node of the heap

Get operation -> Gets the root node
Delete op -> Deletes the root node

IMP

Suppose you need the k^{th} minimum value of min heap ->

Do (k-1) deletions so that k^{th} min value reaches the root & then return the root value

Represent Binary Heap very an Array

Space benefits !

[100, 40, 50, 10, 15, 50, 40]

[6, 3, 2, N, 1, 2, N]

0 1 2 3 4 5 6

[100, 40, 50, 10, 15, 50, 40]

IMP

For a node at index i:
Left child: $2i+1$
Right child: $2i+2$
Parent: $(i-1)/2$

DELETE

STEP-1: Swap the first node with the last node and then delete the last node

STEP-2: Top Down Heapify the root node

Stop since no child of 4 is now greater than 4

[100, 50, 20, 10, 5, 6, 8, 7, 3]

STEP-1

0 1 2 3 4 5 6 7

[3, 50, 20, 10, 5, 6, 8, 7]

idx : 0
left child : 1
right child : 2

while ≥ 1

STEP-2

[50, 10, 20, 7, 5, 6, 8, 3]

Height of a Heap is $O(\log n)$ #

TC of delete: $O(\log n)$ #

INSERT

insert 15

STEP-1:

STEP-2 - Bottom Up Heapify on 15